Code Overview

Main – file that the code is run from. All the methods are used in main, and everything is compiled there for readability and design practice.

ServerConnection – file that contains all methods used for server connection and handles the stream from the server. Put into a separate file for the sake of readability and good design practice. This is also so we do not have to open more than one socket which would increase the likelihood of errors being thrown.

Card – contains the card class used to format the sort each card, put in a separate file for good design practice as well as maintainability.

Lecture Concepts

Enums – used for Rank to be able to compare each rank easily, as Strings do not have order otherwise. An alternative approach would have been to assign UNIQUE = 4 RARE = 3 etc. and compare the numbers that way, however this is a less favourable option from a readability and understanding standpoint.(from Week 1: Enumeration Types and Exceptions)

Try-Catch and Exceptions – used to catch IOExceptions to help with the debugging process, and used try-catch do handle these exception cases. There's no real alternative to using try-catch and exceptions (from Week 1: Enumeration Types and Exceptions)

HashSet – used for the auto-trading system to keep track of cardId for the buy method and keeps track of cardName for the auto-sell method. (from week 4: Inheritance and Interfaces)

ArrayList – used in the cardHelper method used to create the format for the cards when they are displayed. An ArrayList is used here because it can also be sorted easily which is necessary as we want the cards displayed by rank. (from Week 3: Generics, Collections)

Comparable – comparable interface used to compare the different fields in the card class, so the system can automatically compare and sort cards. It does this by using Collections.sort() function from the collection class.(from Week 8: Sorting Algorithms)

Testing Strategy

The methods used to test my code included: testing alterative usernames and passwords to log on the Hollomon server, on the menu providing different inputs to what was expected, for the buy and sell features, providing cardId's that don't exist. I also used System.out.println() to test if particular sections were being run, so I could systematically test the code.

Use of AI

AI used – Claude by Anthropic – ver. Claude Sonnet 4.6 – https://claude.ai

AI was used in this project to help with the debugging process to save time, I have specifically requested that Claude should never give me a coded solution and only ever “nudge me in the right direction”. It was also used to simplify some concepts I found through the Java documentation if I did not fully understand it and could not understand it from the previous lectures.